# Apoorv Srivastava

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## Research Interests

Non-equilibrium dynamics of interacting quantum many-body systems: kinetically constrained and dipole-conserving models, Hilbert-space fragmentation and breakdown of thermalisation, entanglement structure of driven and open systems, and the role of symmetries and conservation laws in relaxation and information retention. I combine analytical many-body methods with exact diagonalisation and open-system numerics, and I am also interested in strongly correlated superconductivity within the extremely correlated (Hubbard-operator) framework.

## Education

**Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram**
Nov, 2020 – May, 2025
*Dual Degree: B.Tech. Engineering Physics and M.S. Solid State Physics • CGPA 8.69/10*

- M.S. thesis: *A Ginzburg–Landau-like theory for high-$T_c$ superconductors within the extremely correlated approach.* Advisor: Prof. T. V. Ramakrishnan (JNCASR / IISc Bangalore).

## Employment

**Scientist/Engineer-'SC', Indian Space Research Organisation (ISRO)** Nov, 2025 – present
*Thin Film Division, Laboratory for Electro-Optical Systems, Bangalore*

- Design and production of Optical Filters for Satellite applications.
- Magnetron Sputtering, Ion-Beam Sputtering, Electron-Beam Evaporation

## Publications and Preprints

### Peer-reviewed

1. **A. Srivastava** and S. Dutta, *Hierarchy of degenerate stationary states in a boundary-driven dipole-conserving spin chain*, SciPost Phys. **18**, 111 (2025).

### In preparation

2. **A. Srivastava**, Y. Sadki, S.L. Sondhi and A. Prakash, *Support-controlled ergodicity in classical continuum fractons*, manuscript in preparation (2026).

## Conference

- *Hierarchy of degenerate stationary states in a boundary-driven dipole-conserving spin chain.* Contributed talk, **Emerging Topics in Quantum Technology**, IIT Palakkad, November 2023.

## Research Experience

**Support-controlled ergodicity in classical continuum fractons** June, 2025 – present
*Dr. Abhishodh Prakash, Department of Physics, Harish-Chandra Research Institute (HRI), Prayagraj*

- Studied ergodicity in one-dimensional classical continuum fractons with dipole conservation and purely relational (Machian) inertia, where isolated particles are strictly immobile and motion arises only within the reach of a pair-inertia kernel.
- Showed that the asymptotic fate is set by the *support* of the kernel rather than by density: compact support retains a nonzero frozen fraction at every density, while any decaying tail reconnects the dynamics and restores ergodicity after long transients. This contrasts with the filling-tuned fragmentation of dipole-conserving lattice chains.

- Found that compactifying the momenta restores ergodicity even for compact support, making Machian clustering an intrinsically continuum effect. Characterised throughout using the frozen fraction as a memory order parameter, mean-squared displacement, scattering functions and single-particle mobility distributions.

**Ginzburg–Landau-like Theory for High-$T_c$ Superconductors — M.S. Thesis**
Jun, 2024 – Oct, 2025
*Prof. T. V. Ramakrishnan, JNCASR and CCMT, IISc Bangalore*
*Co-advisors: Prof. S. R. Hassan (IMSc Chennai) and Prof. N. S. Vidhyadhiraja (JNCASR)*

- Derived a microscopic Ginzburg–Landau-like free-energy functional for strongly correlated superconductors within the extremely correlated Fermi liquid framework, working directly in the Hubbard $X$-operator representation.
- Avoided Gorkov-type factorisation of the order parameter (the average spin-singlet nearest-neighbour pair amplitude), yielding a more general free energy that retains correlation-induced corrections to the gradient and quartic terms.
- Methods: Hubbard operator algebra, Schwinger-source functional techniques, and self-consistent numerical solution of the resulting equations.

**Degenerate Stationary States in a Boundary-Driven Dipole-Conserving Spin Chain**
May 2023 – Dec 2023
*Dr. Shovan Dutta, Theoretical Physics, Raman Research Institute, Bangalore* *Visiting Student Programme*

- Characterised the steady-state manifold of a kinetically constrained, dipole-conserving spin chain subject to incoherent spin drives at its boundaries.
- Showed that the Liouvillian supports a hierarchy of degenerate stationary states, with the number of information-preserving structures growing with system size, so that boundary driving does not erase bulk memory.
- Implemented exact diagonalisation and Lindblad time evolution (Mathematica, Python) to extract spin currents, entanglement entropy and its retention, and to resolve the fragmentation structure of the Hilbert space.
- Published in *SciPost Phys.* **18**, 111 (2025).

## Other Projects

- **Universal properties of quantum Hall edge states** — *reading project, 2024.* Edge velocities and transport for crystallographic and non-crystallographic edge terminations — with the aim to understand if some properties are universal to all directional edges; numerical reproduction of published results.
- **EIT-based quantum memory with squeezed light** — *Dr. Ashok Kumar, IIST, 2023–2024.* Lindblad description of squeezed states propagating through a three-level $\Lambda$ system, using the dark-state polariton picture to identify optimal storage conditions in the presence of noise.
- **Error mitigation for continuous-variable quantum information** — *Dr. Solomon Ivan, IIST, 2022.* Adaptation of Richardson extrapolation from discrete-variable circuits to the continuous-variable setting.
- **Indigenous fluxgate magnetometer and gradiometer for nanosatellites** — *Dr. S. Mukhopadhyay, IIST, 2021–2022; team lead.* Core-material selection and sensor-geometry optimisation via COMSOL; magnetic-signature modelling and MATLAB analysis of sensor time series.

## Technical Skills

**Programming:** Python (proficient), Mathematica (proficient), MATLAB (proficient), Julia and C++ (working knowledge), LaTeX, Git.
**Numerics:** exact diagonalisation, Lindblad and open-system time evolution, DMFT-style self-consistent loops for self-energies and Green's functions, entanglement measures, large-scale symbolic computation. Libraries: NumPy, SciPy, QuTiP, Qiskit, Matplotlib.

**Languages:** Hindi (native), English (C1/C2, medium of instruction throughout),

## Relevant Coursework

**Physics:** Quantum Many-Body Physics, Quantum Mechanics I–II, Advanced Statistical Physics, Statistical Mechanics, Advanced Solid State Physics, Quantum Information Theory, Quantum Optics, Cold Atoms and Bose–Einstein Condensation, Computational Physics, Atomic and Molecular Physics, Mathematical Physics, Classical Mechanics, Electromagnetism and Special Relativity.

**Self-study:** Statistical Field Theory, Non-Equilibrium Many-Body Physics, Tensor Networks, Open Quantum Systems, Quantum Field Theory.

## Fellowships, Awards and Training

- **SERB Fellow**, Department of Science and Technology (DST), at JNCASR Bangalore.
- **Visiting Research Assistant**, Institute of Mathematical Sciences (IMSc), Chennai.
- **Visiting Student Programme**, Raman Research Institute, Bangalore (2023).
- **Department of Space Scholarship**, awarded for all ten semesters of the dual degree.
- Qiskit Global Summer School, IBM Quantum (2022).

## Outreach

- Member, **Nirmaan** (social outreach club, IIST): taught elementary mathematics and science in under-served communities near Thiruvananthapuram; organised blood-donation camps and coastal clean-up drives.

## Referees

**Prof. T. V. Ramakrishnan** — Honorary Professor, Theoretical Sciences Unit, JNCASR Bangalore. *M.Sc. thesis advisor.* `[e-mail]`

**Dr. Shovan Dutta** — Assistant Professor, Raman Research Institute, Bangalore. *Co-author, SciPost Phys. 18, 111 (2025).* `[e-mail]`

**Dr. Abhishodh Prakash** — Reader-F, Harish-Chandra Research Institute, Prayagraj. *Current collaborator.* `abhishodhprakash@hri.res.in`

**Prof. S. R. Hassan** — Institute of Mathematical Sciences, Chennai. *Thesis co-advisor.* `[e-mail]`

**Prof. N. S. Vidhyadhiraja** — Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR), Bangalore *Thesis co-advisor.* `[e-mail]`